

Calculus and Linear Algebra

Sets, Real Numbers and Mathematical Language

Lecture 1

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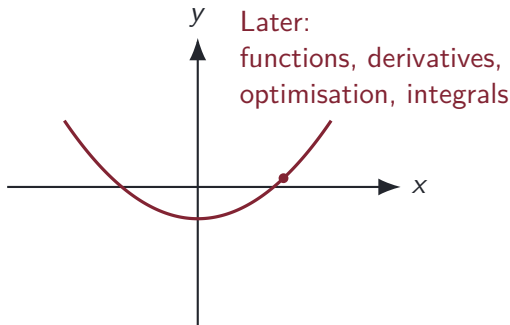
Before we start: what is this course really about?

Learning to read mathematical objects

A formula is not a decoration. It tells us:

- ▶ what objects we are working with;
- ▶ which operations are allowed;
- ▶ what must be true before we calculate;
- ▶ what a result means geometrically.

Today: build the language we will use for the whole course.



Today's route



The point

By the end of the lesson you should be able to translate quickly between **words**, **symbols**, and a **picture on the number line**.

Warm-up: read the symbols before calculating

What do these statements say in ordinary English?

1. $3 \in \mathbb{N}$

2. $-2 \notin \mathbb{N}$

3. $\{1, 2\} \subset \{1, 2, 3\}$

4. $x \in [-1, 4)$

5. $A \cap B = \emptyset$

6. $|x - 2| < 1$

Your turn Do not solve anything yet. Just say what each line means.

A set is a collection of objects

A **set** is a collection whose elements are clearly determined.

$$A = \{1, 3, 5, 7\}$$

We write

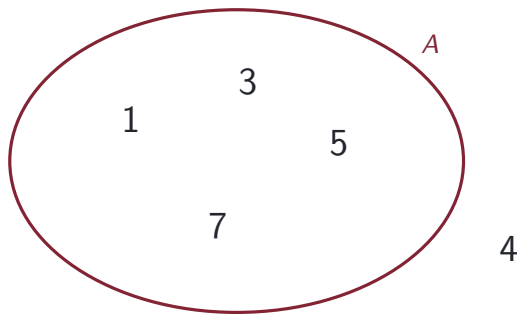
$$3 \in A, \quad 4 \notin A.$$

Order does not matter:

$$\{1, 3, 5\} = \{5, 1, 3\}.$$

Repeated elements do not matter:

$$\{1, 1, 3\} = \{1, 3\}.$$



Two common ways to describe a set

Roster form

List the elements:

$$A = \{2, 4, 6, 8\}.$$

Set-builder form

Describe the property:

$$A = \{x \in \mathbb{N} : 2 \leq x \leq 8, x \text{ even}\}.$$

Good when the set is small and explicit.

Good when the rule matters more than the list.

Your turn Write $\{-3, -2, -1, 0, 1, 2, 3\}$ in set-builder notation.

Exercise: translate into symbols

Write each sentence in mathematical notation.

1. x belongs to the set of real numbers.
2. 5 is not an element of the set A .
3. B contains the elements $-1, 0, 1$.
4. A contains only positive integers smaller than 6.

Notation is a language. Fluency comes from translating both ways.

Subsets: a set inside another set

If every element of A also belongs to B ,
then

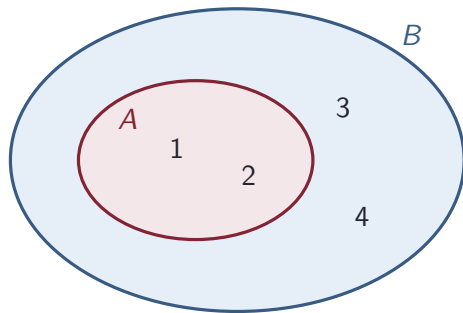
$$A \subseteq B.$$

Example:

$$A = \{1, 2\}, \quad B = \{1, 2, 3, 4\}.$$

Hence $A \subseteq B$. **Important:** an element
and a subset are different objects.

$$1 \in B, \quad \{1\} \subseteq B.$$



A common trap

Let $A = \{1, 2, 3\}$. Which statements are true?

$$1 \in A$$

$$\{1\} \in A$$

$$\{1\} \subseteq A$$

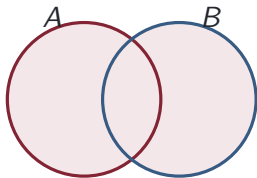
$$A \subseteq A$$

Your turn Vote first. Then justify your answer with words, not only symbols.

Union and intersection

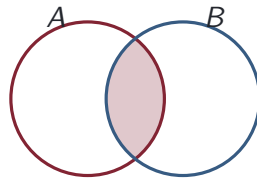
Union: everything in A or B (or both)

$$A \cup B.$$



Intersection: everything in both A and B

$$A \cap B.$$



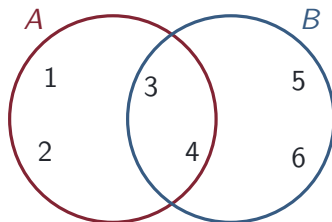
Work from a concrete example

Let

$$A = \{1, 2, 3, 4\}, \quad B = \{3, 4, 5, 6\}.$$

Find:

$$A \cup B, \quad A \cap B, \quad A \setminus B, \quad B \setminus A.$$



Empty set and disjoint sets

The **empty set** has no elements:

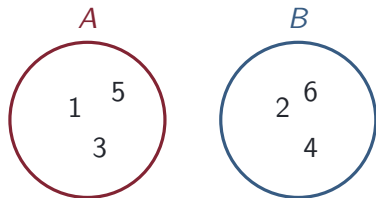
$$\emptyset.$$

Two sets are **disjoint** when they have no common element:

$$A \cap B = \emptyset.$$

Example:

$$\{1, 3, 5\} \cap \{2, 4, 6\} = \emptyset.$$



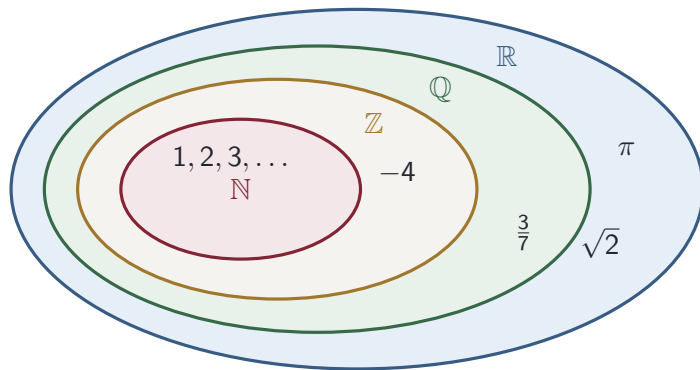
Checkpoint 1

Let $A = \{x \in \mathbb{Z} : -2 \leq x \leq 3\}$ and $B = \{-4, -2, 0, 2, 4\}$.

1. Write A in roster form.
2. Find $A \cap B$.
3. Find $A \cup B$.
4. Is $\{-2, 0, 2\} \subseteq A$?
5. Is $4 \in A \cup B$?

3 minutes – compare answers with the person next to you.

The number system: nested sets



$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}.$$

Natural numbers and integers

Natural numbers

$$\mathbb{N} = \{1, 2, 3, \dots\}$$

Counting numbers.

Some books include 0 in $\mathbb{N}$. Always check the convention.

Integers

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$$

Positive and negative whole numbers, including zero.

Your turn Which set is the smallest one containing each number?

7, -5, 0.

Rational numbers

A number is **rational** if it can be written as a ratio of integers:

$$\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0 \right\}.$$

Examples:

$$\frac{2}{3}, \quad -\frac{11}{4}, \quad 5 = \frac{5}{1}, \quad 0.125 = \frac{1}{8}.$$

A decimal expansion of a rational number either **terminates** or **eventually repeats**.

$$\frac{1}{3} = 0.333\dots, \quad \frac{7}{12} = 0.58333\dots$$

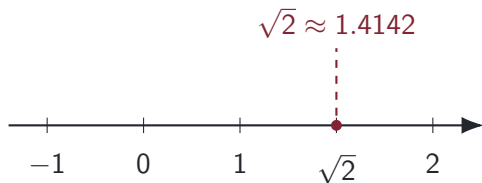
Irrational numbers: real, but not rational

Numbers such as

$$\sqrt{2}, \quad \pi, \quad e$$

cannot be written as p/q with integers p, q .
They are **irrational**. Together,

$$\text{rationals} + \text{irrationals} = \mathbb{R}.$$

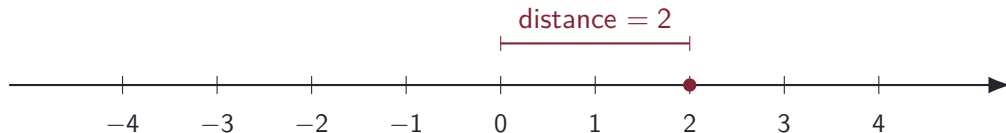


Your turn: classify the numbers

For each number, choose the **smallest** set among $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ that contains it.

Number	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$
12				
-3				
$\frac{7}{5}$				
0.4				
$\sqrt{9}$				
$\sqrt{5}$				

The real line: numbers become geometry



Key idea

The order relation $<$ gives the line a direction. Distance turns algebra into geometry.

Your turn Where would you place -2.5 , $1/3$, and $\sqrt{2}$?

Intervals: a compact way to describe many numbers

Closed interval

$$[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$$



Open interval

$$(a, b) = \{x \in \mathbb{R} : a < x < b\}$$



Square bracket = endpoint included. Round bracket = endpoint excluded.

Half-open intervals

$$[a, b) = \{x : a \leq x < b\}$$

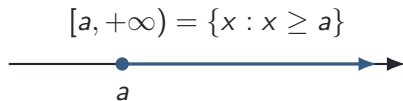
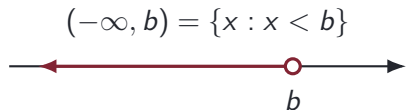


$$(a, b] = \{x : a < x \leq b\}$$



Your turn Draw $[-2, 3)$ on a number line.

Unbounded intervals



$\pm\infty$ are not real numbers, so we never use square brackets with them.

Fast translation drill

Translate each interval into an inequality, then sketch it.

1. $[-3, 2]$

2. $(1, 5)$

3. $(-\infty, 4]$

4. $[-1, +\infty)$

5. $(-2, 0]$

6. $\mathbb{R}$

Goal: interval notation should become automatic.

And now the reverse translation

Write each inequality using interval notation.

1. $-2 \leq x < 4$

2. $x > 3$

3. $x \leq -1$

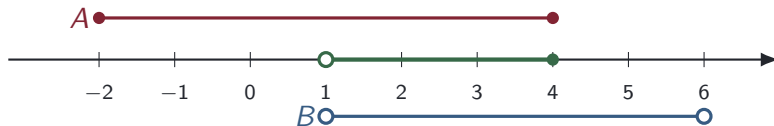
4. $0 < x \leq 7$

Your turn For each one, mark endpoints with open/closed circles before writing the interval.

Intervals are sets: we can intersect them

Find

$$[-2, 4] \cap (1, 6).$$

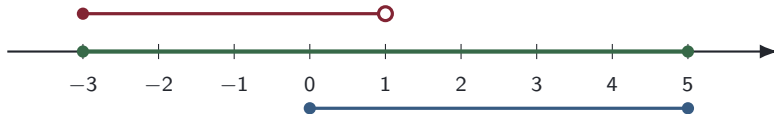


$$[-2, 4] \cap (1, 6) = \boxed{(1, 4]}.$$

Intervals are sets: and we can take unions

Find

$$[-3, 1) \cup [0, 5].$$



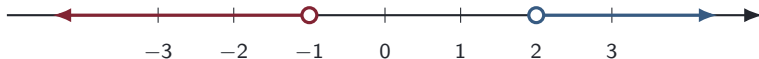
$$[-3, 1) \cup [0, 5] = \boxed{[-3, 5]}.$$

The two intervals overlap, so the union is one continuous interval.

What if there is a gap?

Find

$$(-\infty, -1) \cup (2, +\infty).$$



Can this union be written as a single interval? Why not?

Inequalities: think on the line

Consider

$$2x - 3 < 5.$$

Solving is not just algebra; it identifies a set of real numbers:

$$2x < 8 \implies x < 4.$$



$$\boxed{(-\infty, 4)}.$$

One rule you must not forget

When you multiply or divide an inequality by a **negative number**, the inequality sign reverses.

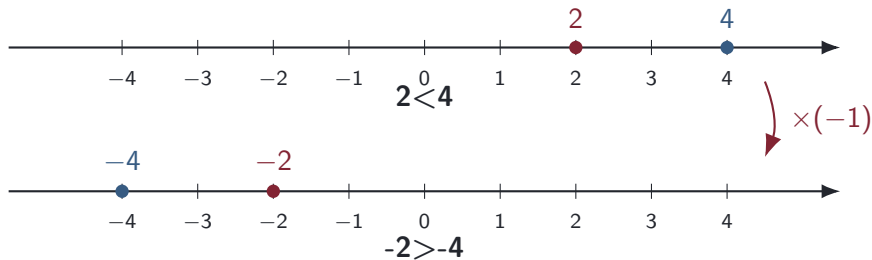
$$-2x > 6$$

↓ divide by -2

$$\boxed{x < -3}.$$

Why? Because multiplication by -1 reflects the number line around zero.

See the sign reversal geometrically



Exercise set: linear inequalities

Solve and write the solution in interval notation.

1. $3x + 1 \leq 10$

2. $5 - 2x > 1$

3. $4x - 7 \geq 2x + 3$

4. $2(1 - x) < 3x + 7$

5. $-3 \leq 2x + 1 < 7$

For no. 5, treat the double inequality as two conditions at once.

Double inequalities

Solve

$$-3 \leq 2x + 1 < 7.$$

We perform the same operation on all three parts:

$$-4 \leq 2x < 6$$

$$-2 \leq x < 3.$$



$$\boxed{[-2, 3)}.$$

Absolute value is distance from zero

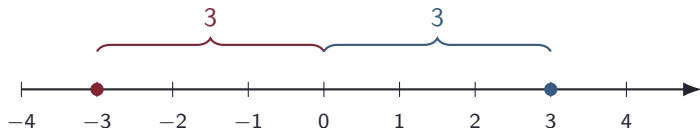
$|x|$ = distance of x from 0.

Therefore

$$|3| = 3, \quad |-3| = 3.$$

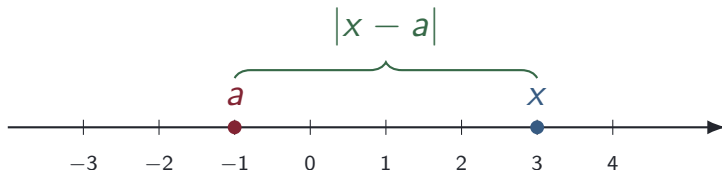
Distance is never negative:

$$|x| \geq 0.$$



Absolute value between two points

$|x - a|$ = distance between x and a .



This interpretation will reappear in limits, continuity, approximation and optimisation.

Absolute-value inequalities: start from the picture

What does

$$|x| < 2$$

mean? It says: *the distance of x from zero is less than 2.*



$$-2 < x < 2$$

$\iff$

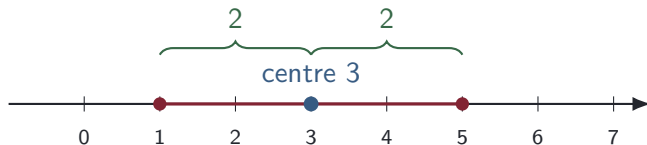
$$x \in (-2, 2).$$

Shift the centre

Solve

$$|x - 3| \leq 2.$$

Interpretation: x must be at distance at most 2 from the centre 3.



$$\boxed{1 \leq x \leq 5}.$$

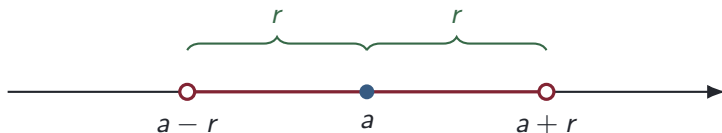
General rule: inside a radius

For $r > 0$,

$$|x - a| < r$$

means that x lies **inside** the interval of radius r centred at a :

$$a - r < x < a + r.$$



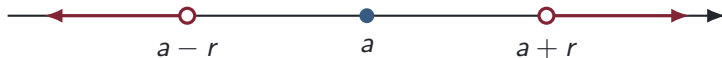
General rule: outside a radius

For $r > 0$,

$$|x - a| > r$$

means that x lies **outside** the interval of radius r centred at a :

$$x < a - r \quad \text{or} \quad x > a + r.$$



Exercise set: absolute value

Solve and represent the solution on the real line.

1. $|x| < 4$

2. $|x| \geq 3$

3. $|x - 2| < 1$

4. $|x + 1| \leq 2$

5. $|2x - 1| < 5$

Always ask first: what is the centre? what is the radius?

From algebra to geometry: $|2x - 1| < 5$

$$|2x - 1| < 5$$

$$-5 < 2x - 1 < 5$$

$$-4 < 2x < 6$$

$$\boxed{-2 < x < 3}.$$



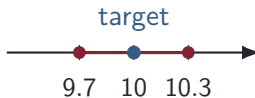
A useful application: tolerance

Suppose a measurement is intended to be 10, with tolerance 0.3. “Acceptable” means

$$|x - 10| \leq 0.3.$$

Therefore

$$9.7 \leq x \leq 10.3.$$



A preview of functions: domain restrictions come from sets

Soon we will study functions such as

$$f(x) = \frac{1}{x-2}, \quad g(x) = \sqrt{x+1}, \quad h(x) = \log x.$$

Before doing anything else, we must ask:

For which real numbers is the formula meaningful?

$$x \neq 2$$

$$x \geq -1$$

$$x > 0$$

Today's set and interval notation will become the language of **domains**.

Challenge: describe the domain as a set

Write the natural domain of each expression in interval notation.

1. $f(x) = \frac{1}{x+4}$

2. $g(x) = \sqrt{5-x}$

3. $h(x) = \frac{1}{\sqrt{x-1}}$

This is only a preview. The point is to use intervals correctly.

Exercises

1. Write $A = \{x \in \mathbb{Z} : -1 < x \leq 4\}$ in roster form.
2. Compute $[-2, 3) \cap [1, 5]$.
3. Compute $(-\infty, 0] \cup (-1, 2)$.
4. Solve $3 - 2x \leq 7$.
5. Solve $-1 < 2x + 3 \leq 7$.
6. Solve $|x - 4| < 2$.
7. Solve $|x + 2| \geq 3$.
8. Classify $-7/2$, $\sqrt{16}$ and π .

8 minutes – write full notation, not only final numbers.

Quick solutions

1. $A = \{0, 1, 2, 3, 4\}$

2. $[1, 3)$

3. $(-\infty, 2)$

4. $x \geq -2 \Rightarrow [-2, +\infty)$

5. $-2 < x \leq 2 \Rightarrow (-2, 2]$

6. $2 < x < 6 \Rightarrow (2, 6)$

7. $x \leq -5$ or $x \geq 1$

8. $-7/2 \in \mathbb{Q}$, $\sqrt{16} = 4 \in \mathbb{N}$, $\pi \in \mathbb{R} \setminus \mathbb{Q}$

Check Compare your reasoning, not only the answers.

Five ideas to leave with today

1. A set is a collection of well-defined objects.
2. $\in$ relates an **element** to a set; $\subseteq$ relates one **set** to another.
3. $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$.
4. Intervals are sets of real numbers described by inequalities.
5. Absolute value is distance: $|x - a|$ measures how far x is from a .

The real line should now be your default mental picture.

Exit ticket

Before you leave, answer these three questions without notes.

1. What is the difference between $2 \in A$ and $\{2\} \subseteq A$?
2. Write $-1 \leq x < 3$ as an interval.
3. Solve $|x - 5| < 2$ without expanding anything first.

Next lecture: **Vectors – geometry, coordinates and operations.**

**Mathematics becomes easier
when the picture and the
symbols
say the same thing.**

See you on Friday.
